Module 12, Lesson 2

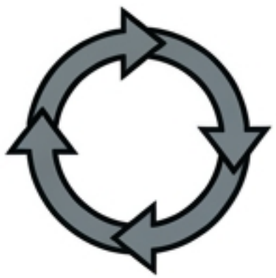
AI 101 for Mechanical Engineers

Sequence architectures are different mechanisms



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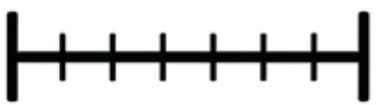
Lesson Sections



1. Sequence Architectures



2. Architecture Interpretation



3. Forecasting Constraints

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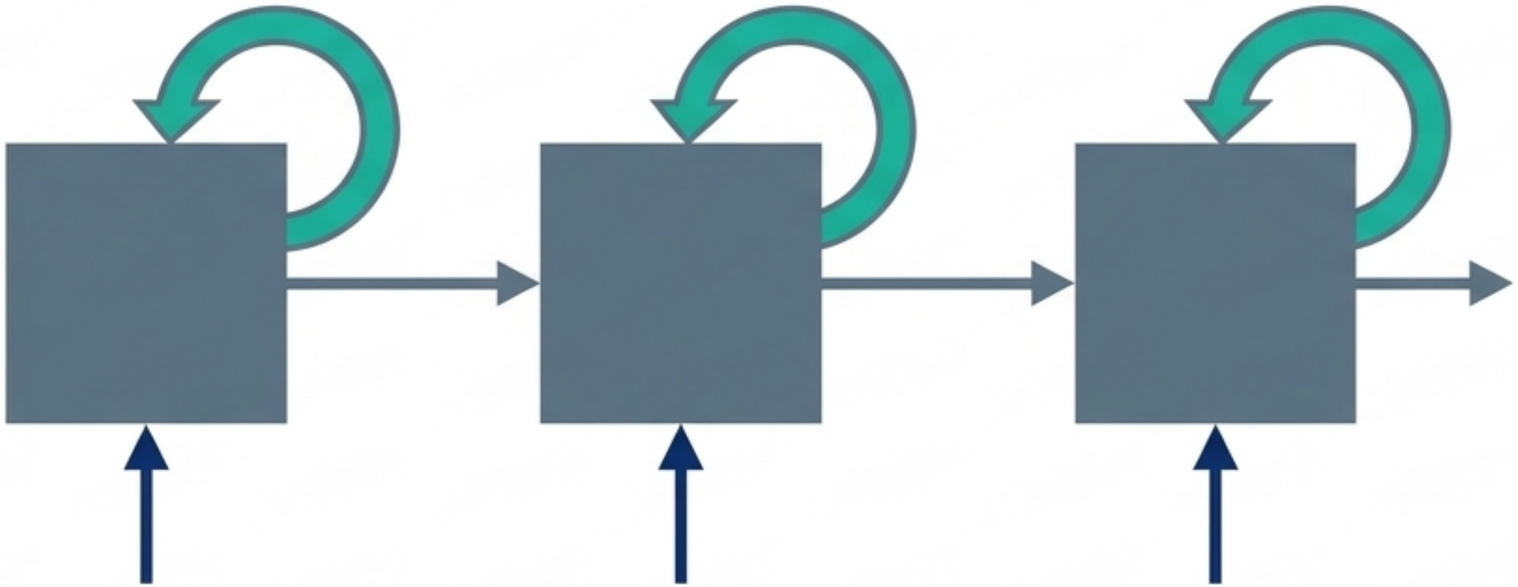
Section 1: Sequence Architectures

Distinguish recurrence, LSTM, and attention mechanisms.

[1. Sequence Architectures | 2. Architecture Interpretation | 3. Forecasting Constraints]

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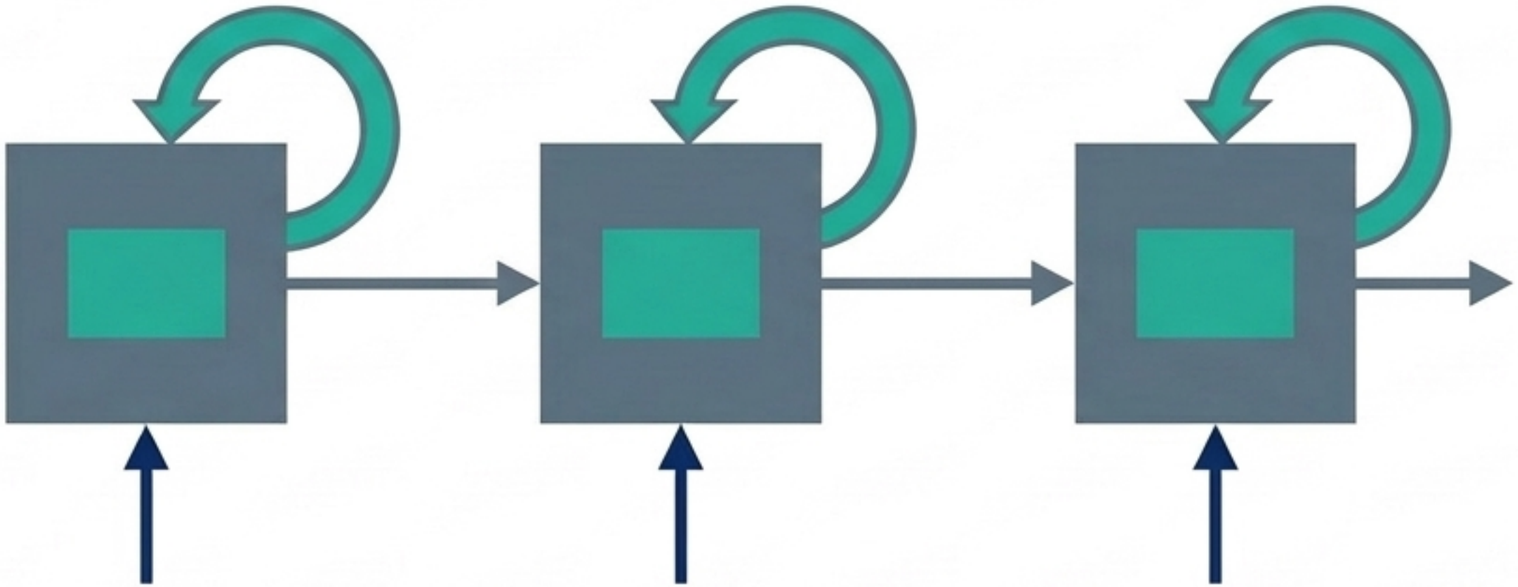
State Propagation in RNNs



RNNs carry a hidden state through sequential steps.	Hidden state updates sequentially with each data input.	Applicable for processing continuous sequences of sensor values.

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Information Management in LSTMs



LSTMs represent a specific recurrent architecture design.	Internal mechanisms retain and update information over time.	Functionally designed to model complex temporal relationships effectively.

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Attention Mechanisms in Transformers

Transformers process sequences using distinct mathematical attention mechanisms.

Attention structures weight and combine representations across positions.

Architecture requires explicit structural representation of sequence order.



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Summary of Information Handling

Architecture	Core Mechanism	Primary Information Handling
RNN	Recurrence	Hidden state through sequence steps
LSTM	Internal Memory	Recurrent information management
Transformer	Attention	Attention combined across positions

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Section 2: Architecture Interpretation

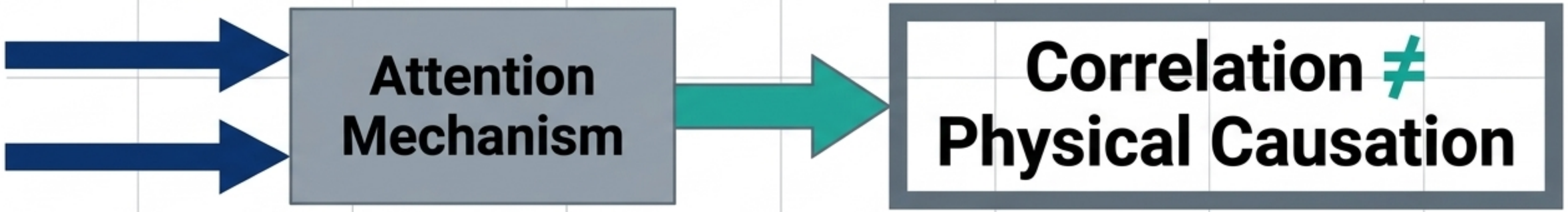
Establish operational realities and model selection criteria.

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Interpreting Attention Weights

Attention structurally weights contributions from distinct data representations.

Mathematical weights do not equate to human attention.



Attention weights do not prove physical causal importance.

Weight distributions alone cannot validate overall model correctness.

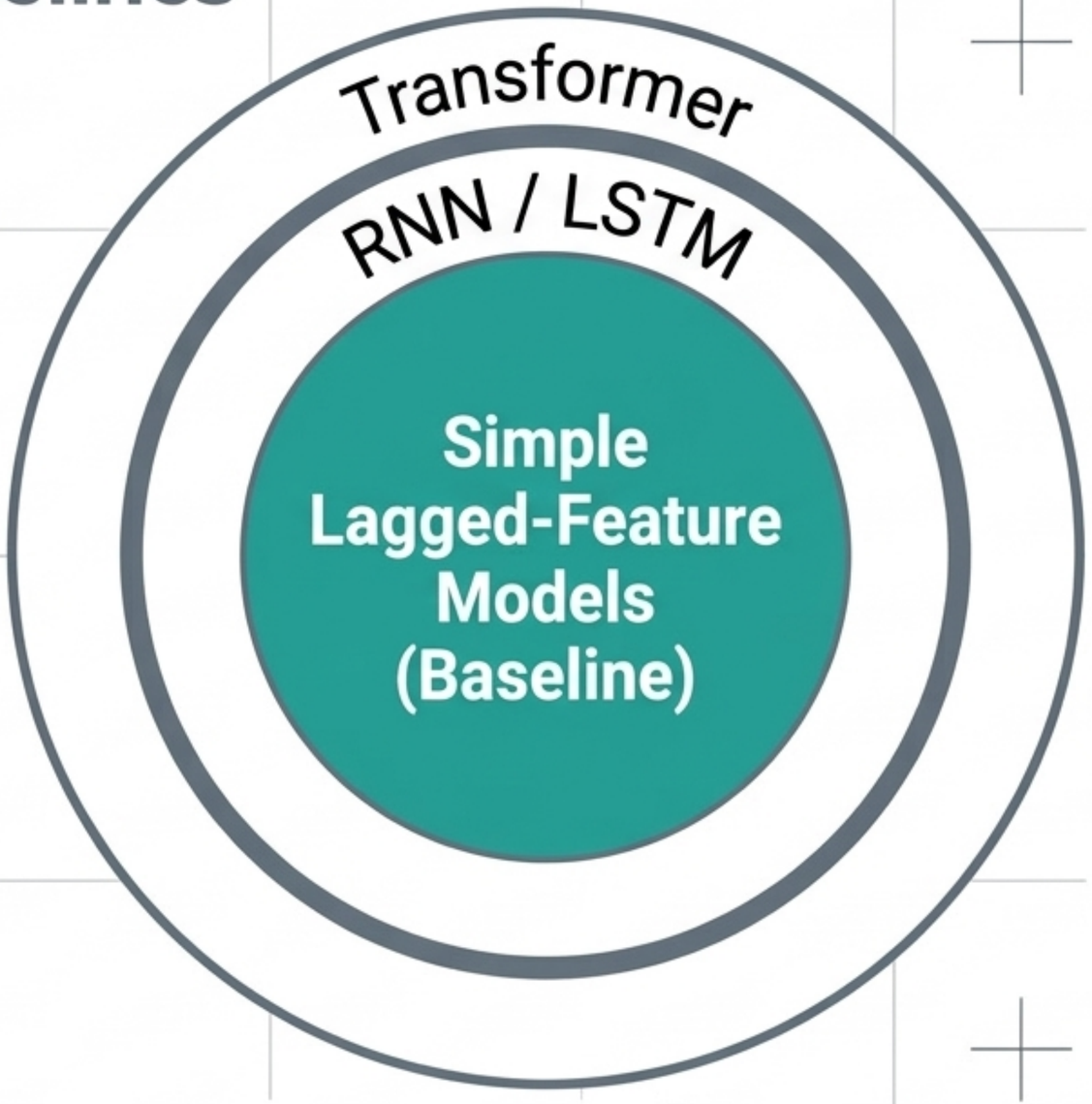
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Establishing Forecasting Baselines

Optimal forecasting methods depend strictly on data constraints.

Transformers and recurrent networks utilize entirely distinct mechanisms.

Simple lagged-feature models often provide strong operational baselines.



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Section 3: Forecasting Constraints

Define rigid information availability boundaries at forecast issue time.

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Information Availability Boundaries



Load prediction requires a precisely defined issue time.

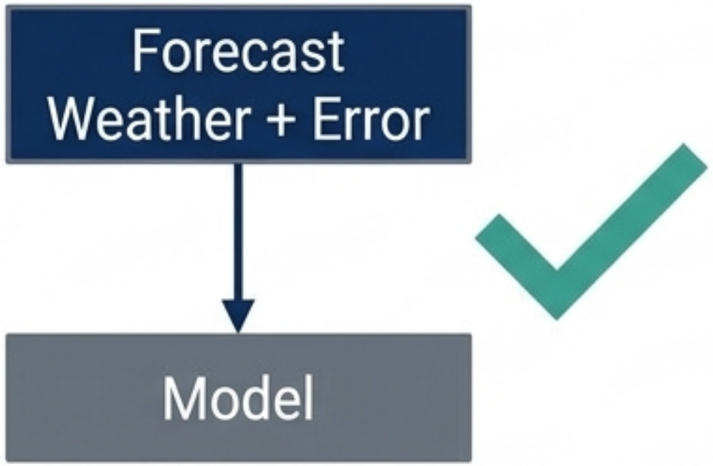
Forecast input variables must must explicitly exist at issue.

Tomorrow's actual measured temperature remains strictly unavailable today.

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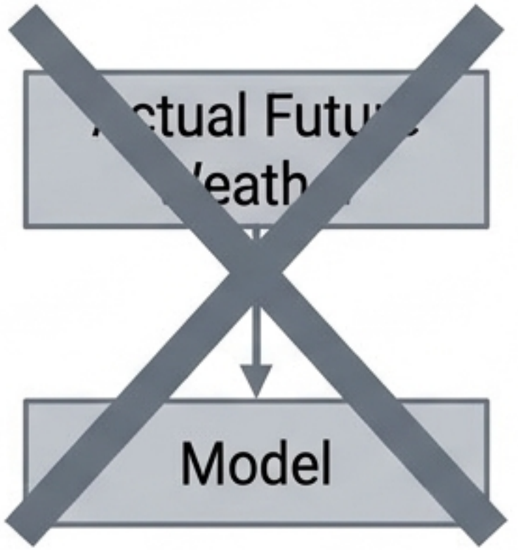
Accounting for Forecast Error

Operational Reality



Weather forecasts available at issue contain inherent errors.

Invalid Evaluation



Measured-weather accuracy cannot represent operational forecast accuracy. Evaluation strictly requires data functionally available at issue.

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Key Points

1. RNNs utilize recurrence; Transformers utilize **attention** across positions.
2. Attention mechanisms structurally do not **prove physical causation**.
3. Operational models **strictly require inputs available at issue**.

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